

Laboration 2:

The role of solid boundaries:

Evolution of an initial disturbance

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Aim

The aim of this assignment is to gain understanding of the influence solid boundaries have on geophysical flows. Although applicable to some situations in the atmosphere, the influence of lateral boundaries is more important in the world oceans. In this assignment you will use a simple shallow water model with solid boundaries. You will test two different initial conditions; an initial disturbance and a system initially at rest with added forcing, and look at how they evolve in time.

Part 1 - Gaussian disturbance

Model setup

- Program an infinite coast by setting an open boundary in the north side of the domain and solid boundary in the south side. Impose periodic boundary conditions in the x-direction.
- Program a new initial condition that describes a Gaussian disturbance of reasonable size of the h -field centered in the middle of the coast, so that the disturbance is "cut" by the wall.
- Introduce reduced gravity to the system.

Experiment

- Run the model for two different cases; (1) the North Atlantic ($L = 1 \cdot 10^7$ m , $H = 1000$ m) and (2) the Baltic Sea ($L = 1 \cdot 10^6$ m , $H = 30$ m).
- Run the model again for both cases (North Atlantic and the Baltic Sea) using different settings, e.g. changing resolution.

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Part 2 - Equatorial beta-plane

In this part of the assignment you are tasked to study waves that appear on an equatorial β -plane.

Model setup

- Update the boundary conditions to be open at both the north and south sides of the domain, while keeping periodic boundary conditions in the x-direction.
- Change the initial condition to a Gaussian disturbance programmed in the middle of the domain.
- Implement a new Coriolis parameter corresponding to an equatorial β -plane ($f = \beta y$). Use a LOGICAL switch to allow easy selection between a β -plane and an f -plane, which will help control the code. Insert this where the Coriolis term is added to the governing equations.
- Remember to place the equator ($y = 0$) in the middle of your domain. This means shifting the coordinates vy_t and vy_v .
- This requires reduced gravity in the code.

Experiment

- Run the model using values responding to the Pacific Ocean ($L = 2 \cdot 10^7$ m , $H = 1000$ m)

Good luck!

and remember, if you're stuck with the programming, ask for help by your classmates or by the assistant.